

WEEK12

Q) A binary number is a combination of 1s and 0s. Its n^{th} least significant digit is the n^{th} digit starting from the right starting with 1. Given a decimal number, convert it to binary and determine the value of the 4th least significant digit.

Example:

number=23

- Convert the decimal number 23 to binary number: $23^{10} = 2^4 + 2^2 + 2^1 + 2^0 = (10111)_2$.
- The value of the 4th index from the right in the binary representation is 0.

Function Description

Complete the function `fourthBit` in the editor below. `fourthBit`

has the following parameter(s):

`int number`: a decimal integer Returns:

`int`: an integer 0 or 1 matching the 4th least significant digit in the binary representation of `number`.

Constraints

$$0 \leq \text{number} < 2^{31}$$

Input Format for Custom Testing

Input from `stdin` will be processed as follows and passed to the function. The only line contains an integer, `number`.

Sample Case 0

Sample Input 0

STDIN Function

32 → `number` = 32

Sample Output 0

0

Explanation0

- Convert the decimal number 32 to binary number: $32_{10} = (100000)_2$.
- The value of the 4th index from the right in the binary representation is 0.

SampleCase1**SampleInput1**

77 → number = 77

SampleOutput1

1

Explanation1

- Convert the decimal number 77 to binary number: $77_{10} = (1001101)_2$.
- The value of the 4th index from the right in the binary representation is 1.

```

1  /*
2   * Complete the 'fourthBit' function below.
3   *
4   * The function is expected to return an INTEGER.
5   * The function accepts INTEGER number as parameter.
6   */
7
8  int fourthBit(int number)
9  {
10     int bin[32];
11     int i=0;
12     while(number>0){
13         bin[i]=number%2;
14         number/=2;
15         i++;
16     }
17     if(i>=4){
18         return bin[3];
19     }
20     else
21         return 0;
22 }

```

	Test	Expected	Got	
✓	printf("%d", fourthBit(32))	0	0	✓
✓	printf("%d", fourthBit(77))	1	1	✓

Passed all tests! ✓

Q)Determinethefactorsof anumber(i.e.,allpositiveintegervalueshatevenlydivideinto a number) and then return the p^{th} element of the list, sorted ascending. If there is no p^{th} element,return 0.

Example

$n = 20$

$p = 3$

Thefactors of20inascending orderare{1,2, 4, 5,10,20}.Using 1-basedindexing,if $p=3$, then 4 is returned. If $p > 6$, 0 would be returned.

FunctionDescription

ComplethefunctionpthFactorintheeditorbelow. pthFactor

has the following parameter(s):

intn:theintegerwhosefactorsaretobefound int

p:the index of the factor to be returned

Returns:

int:thelongintegervalueof the p^{th} integerfactorofn,ifthereisnofactoratthatindex, then 0 is returned

Constraints

$$1 \leq n \leq 10^{15}$$

$$1 \leq p \leq 10^9$$

InputFormatforCustomTesting

Inputfromstdinwillbeprocessedasfollowsandpassedto thefunction. The first line contains an integer n , the number to factor.

Thesecondlinecontainsaninteger p ,the1-basedindexofthefactortoreturn.

SampleCase0

SampleInput0

10 →n= 10

3 →p= 3

SampleOutput0

5

Explanation0

Factoringn=10resultsin{1,2,5,10}.Returnthep=3rdfactor,5,astheanswer.

SampleCase1**SampleInput1**

STDIN Function

10 →n= 10

5 →p= 5

SampleOutput1

0

Explanation1

Factoringn=10resultsin {1,2,5,10}.Thereareonly4factors andp=5, therefore0is returned as the answer.

SampleCase2**SampleInput2**

1 →n= 1

1 →p= 1

SampleOutput2

1

Explanation2

Factoringn=1resultsin{1}.Thep=1stfactorof1isreturnedasthe answer.

```

1  /*
2  * Complete the 'pthFactor' function below.
3  *
4  * The function is expected to return a LONG_INTEGER.
5  * The function accepts following parameters:
6  * 1. LONG_INTEGER n
7  * 2. LONG_INTEGER p
8  */
9
10 long pthFactor(long n, long p)
11 {
12     int c=0;
13     for(long i=1;i<=n;++i){
14         if(n%i==0){
15             c++;
16             if(c==p){
17                 return i;
18             }
19         }
20     }
21     return 0;
22 }
23

```

Test	Expected	Got	
✓ printf("%ld", pthFactor(10, 3))	5	5	✓
✓ printf("%ld", pthFactor(10, 5))	0	0	✓
✓ printf("%ld", pthFactor(1, 1))	1	1	✓

Passed all tests! ✓

Q) You are a bank account hacker. Initially you have 1 rupee in your account, and you want exactly N rupees in your account. You wrote two hacks, first hack can multiply the amount of money you own by 10, while the second can multiply it by 20. These hacks can be used any number of times. Can you achieve the desired amount N using these hacks.

Constraints:

$$1 \leq T \leq 100$$

$$1 \leq N \leq 10^{12}$$

Input

- The test case contains a single integer N .

Output

For each test case, print a single line containing the string "1" if you can make exactly N rupees or "0" otherwise.

SAMPLEINPUT

1

SAMPLEOUTPUT

1

SAMPLEINPUT

2

SAMPLEOUTPUT

0

```
1  /*
2   * Complete the 'myFunc' function below.
3   *
4   * The function is expected to return an INTEGER.
5   * The function accepts INTEGER n as parameter.
6   */
7
8  int myFunc(int n)
9  {
10     return n%10==0 || n==1;
11 }
12
```

	Test	Expected	Got	
✓	printf("%d", myFunc(1))	1	1	✓
✓	printf("%d", myFunc(2))	0	0	✓
✓	printf("%d", myFunc(10))	1	1	✓
✓	printf("%d", myFunc(25))	0	0	✓
✓	printf("%d", myFunc(200))	1	1	✓

Passed all tests! ✓

Q) Find the number of ways that a given integer, X , can be expressed as the sum of the N^{th} powers of unique, natural numbers.

For example, if $X=13$ and $N=2$, we have to find all combinations of unique squares adding up to 13 . The only solution is $2^2 + 3^2$.

Function Description

Complete the powerSum function in the editor below. It should return an integer that represents the number of possible combinations.

powerSum has the following parameter(s):

X : the integer to sum to

N : the integer power to raise numbers to Input

Format

The first line contains an integer X .

The second line contains an integer N .

Constraints

$$1 \leq X \leq 1000$$

$$2 \leq N \leq 10$$

Output Format

Output a single integer, the number of possible combinations calculated.

Sample Input 0

10

2

Sample Output 0

1

Explanation 0

If $X=10$ and $N=2$, we need to find the number of ways that 10 can be represented as the sum of squares of unique numbers.

$$10 = 1^2 + 3^2$$

This is the only way in which 10 can be expressed as the sum of unique squares.

Sample Input 1

100

2

SampleOutput1

3

Explanation1

$100 = (10^2) = (6^2 + 8^2) = (1^2 + 3^2 + 4^2 + 5^2 + 7^2)$

SampleInput2

100

3

SampleOutput2

1

Explanation2

100 can be expressed as the sum of the cubes of 1, 2, 3, 4.

(1 + 8 + 27 + 64 = 100). There is no other way to express 100 as the sum of cubes.

```
1  /*
2  * Complete the 'powerSum' function below.
3  *
4  * The function is expected to return an INTEGER.
5  * The function accepts following parameters:
6  * 1. INTEGER x
7  * 2. INTEGER n
8  */
9
10 int powerSum(int x, int m, int n)
11 {
12     int power = m;
13     for(int i=1; i<n; i++){
14         power*=m;
15     }
16     if(power > x){
17         return 0;
18     }
19     if(power == x){
20         return 1;
21     }
22     return powerSum(x-power,m+1,n) + powerSum(x,m+1,n);
23
24 }
```

Test	Expected	Got	
✓ printf("%d", powerSum(10, 1, 2))	1	1	✓

Passed all tests! ✓